INCH-POUND
MIL-DTL-24368/2C(NAVY)
27 July 2021
SUPERSEDING
MIL-C-24368/2B(NAVY)
25 March 1987

DETAIL SPECIFICATION SHEET

CONNECTOR ASSEMBLIES; RECEPTACLE AND RECEPTACLE–CABLED, POWER TRANSFER, SHORE TO SHIP AND SHIP TO SHIP, 500 VOLTS, 500 AMPERES, 60 HERTZ, SYMBOL NUMBER 1161

This specification is approved for use by the Department of the Navy and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-C-24368.

All dimensions are in inches.

For tolerances other than those specified, see the dimensions and tolerances of MIL-C-24368.

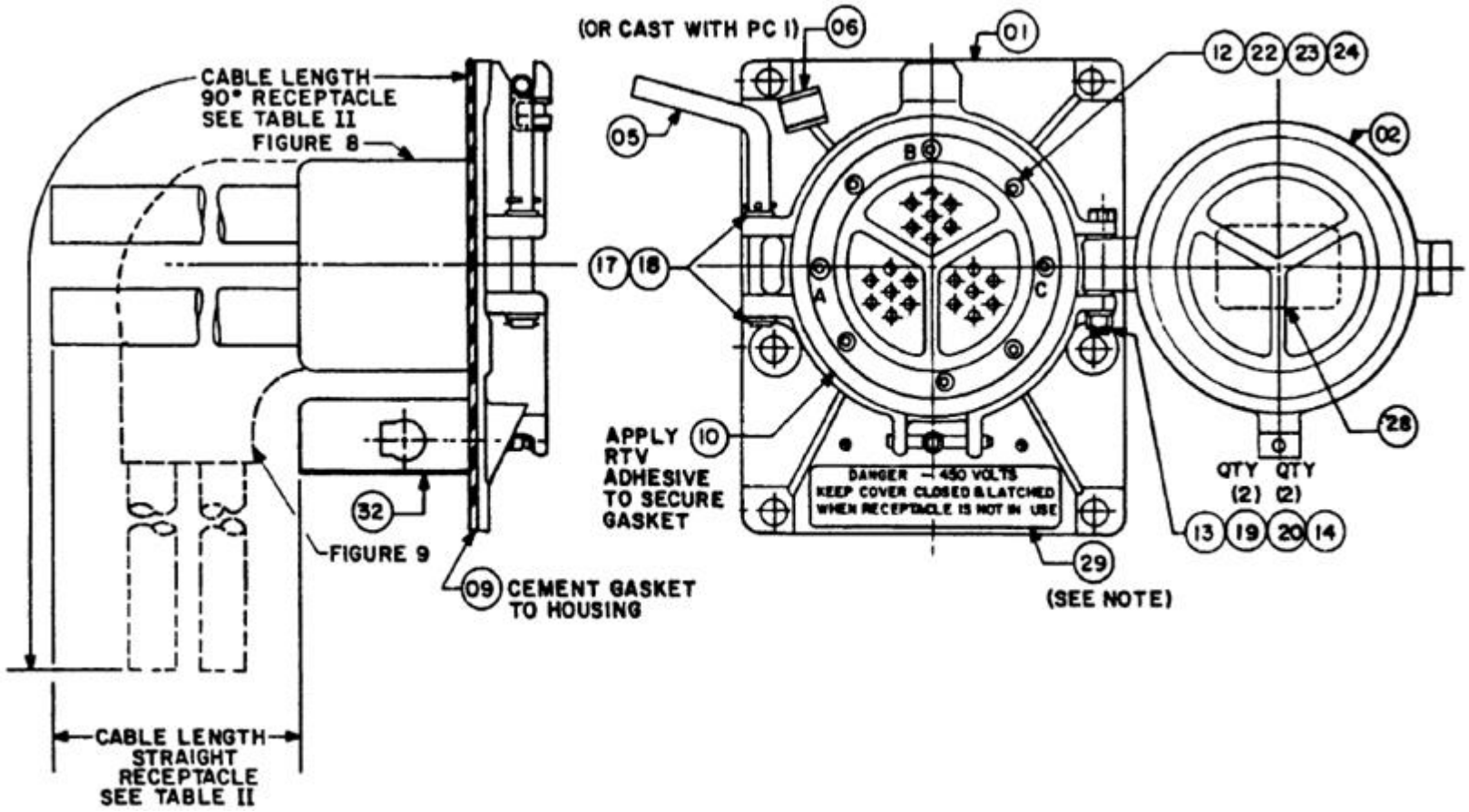


FIGURE 1. Power transfer receptacle, typical.

AMSC N/A FSC 6150

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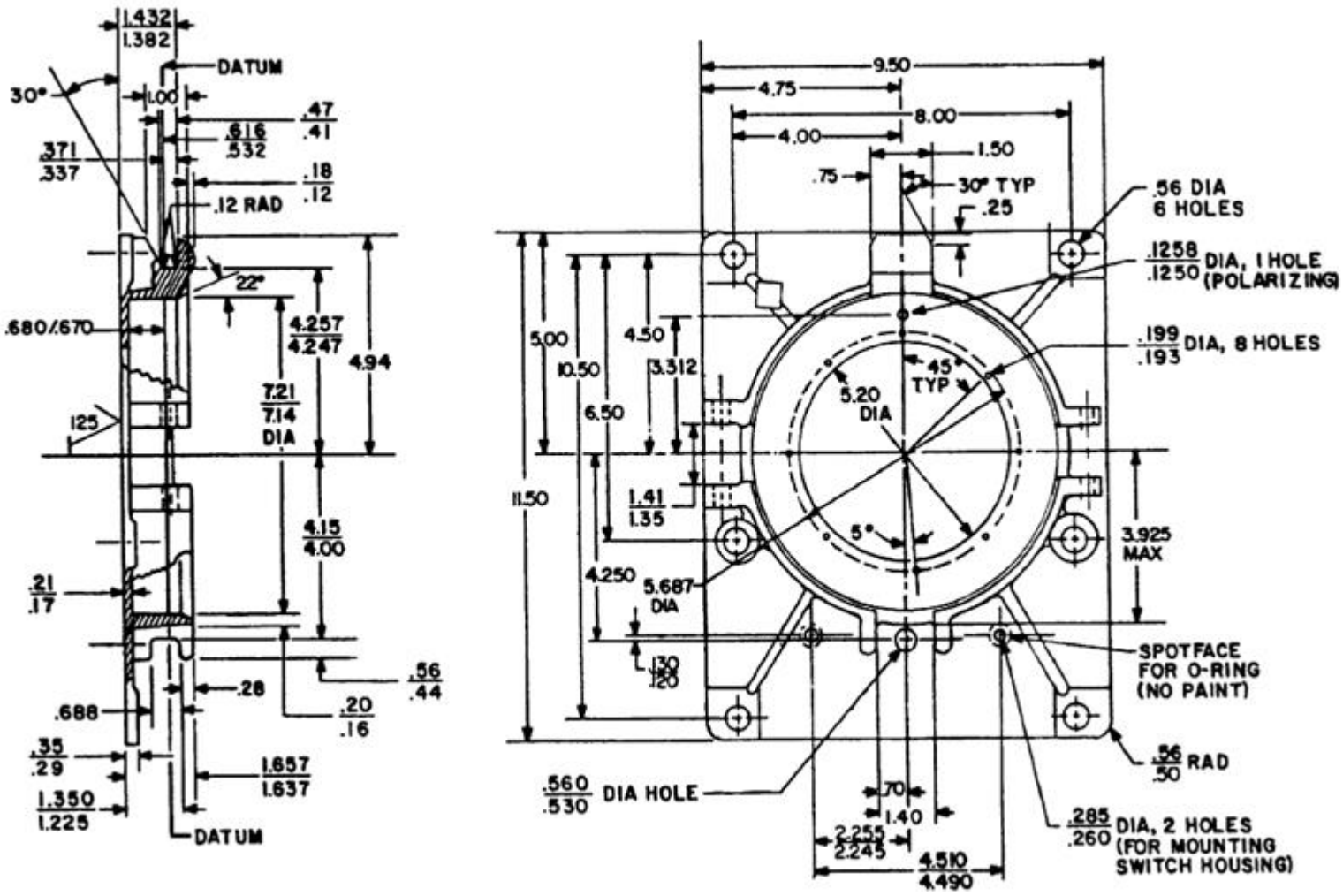
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NOTE: Letters shall be white on red background. Voltage numbers shall be black on white background.

FIGURE 2. Receptacle, typical.

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NOTE: Housing shall be cast as shown.

FIGURE 3. Receptacle housing. (01)

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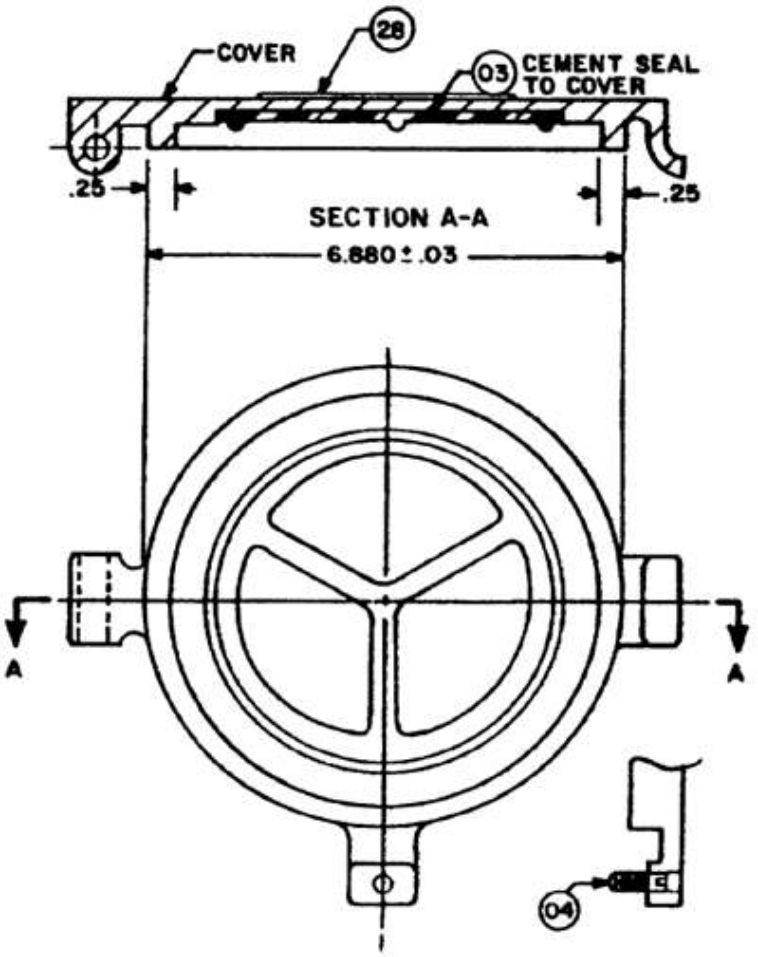


FIGURE 4. Cover assembly. (02)

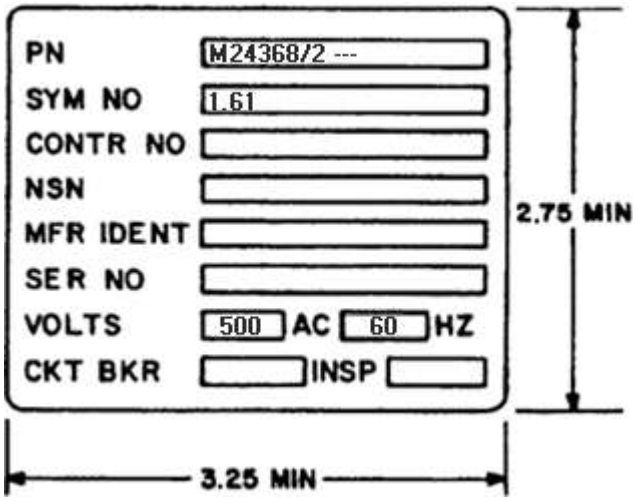


FIGURE 5. Identification plate. (28)

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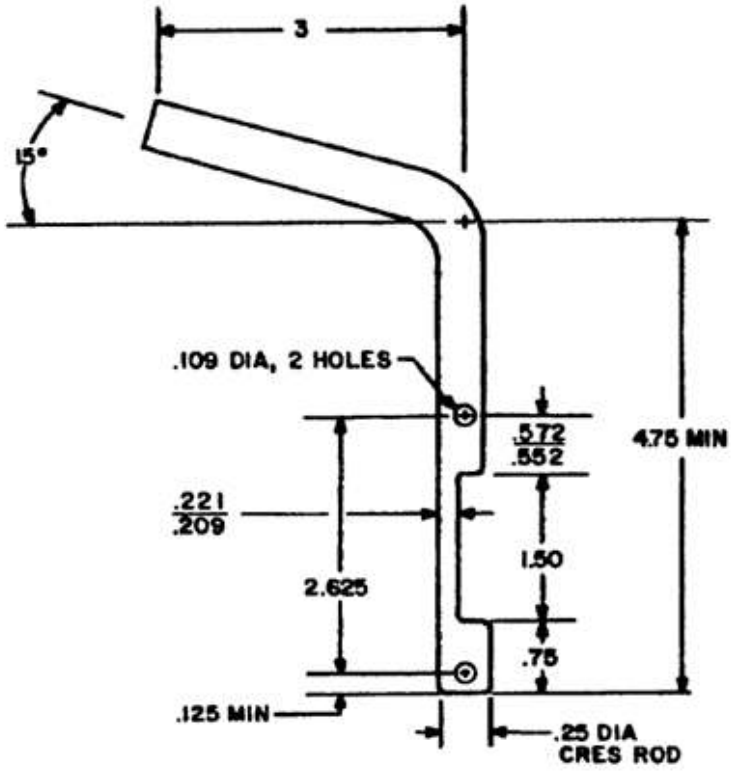
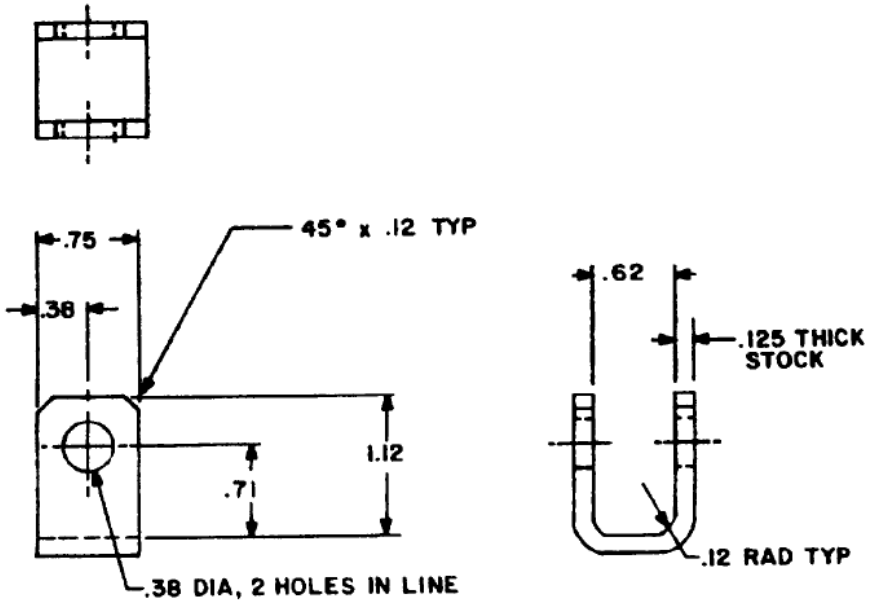


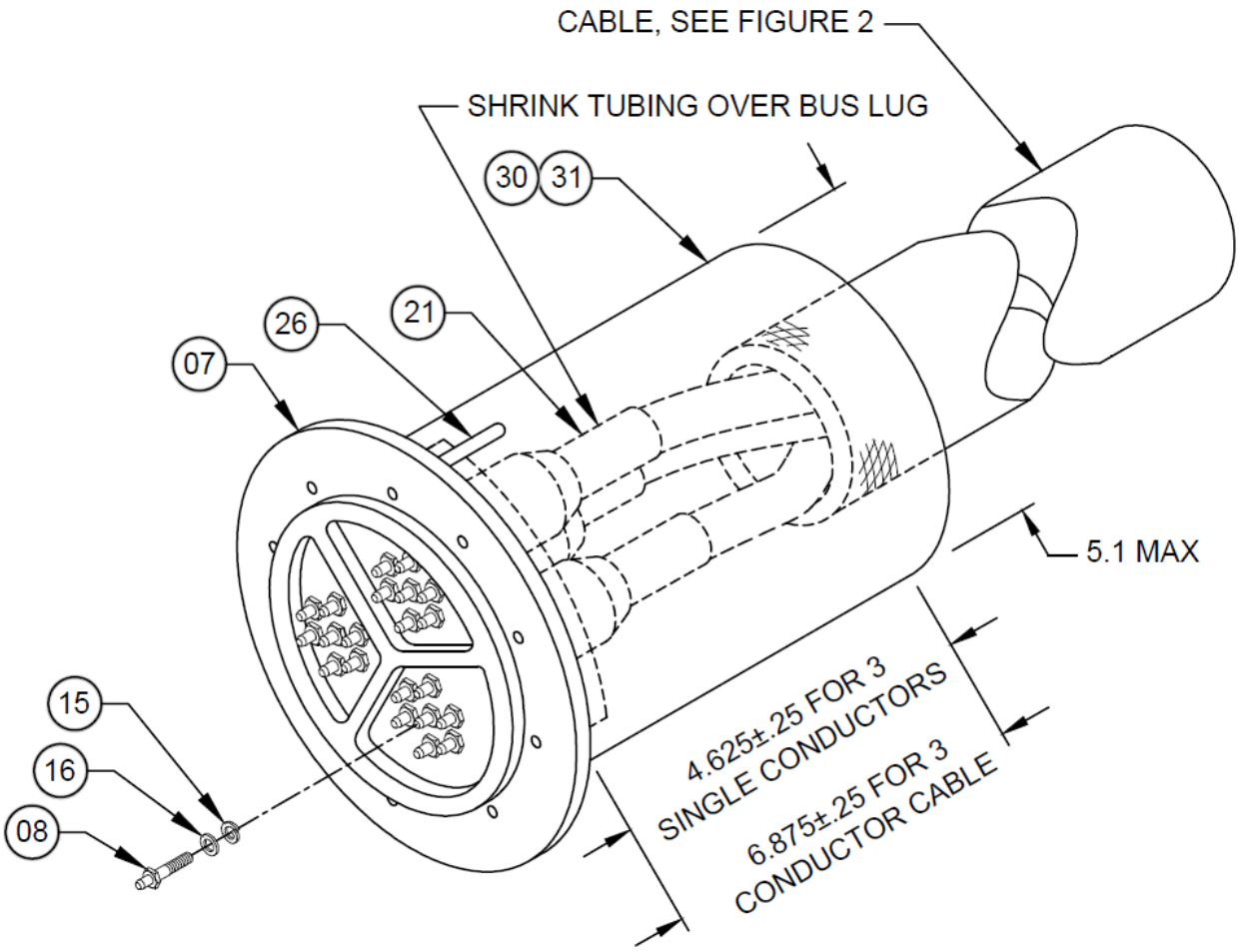
FIGURE 6. Latch. (05)



NOTE: Bracket may be cast with receptacle housing at manufacturer's option.

FIGURE 7. Latch locking bracket. (06)

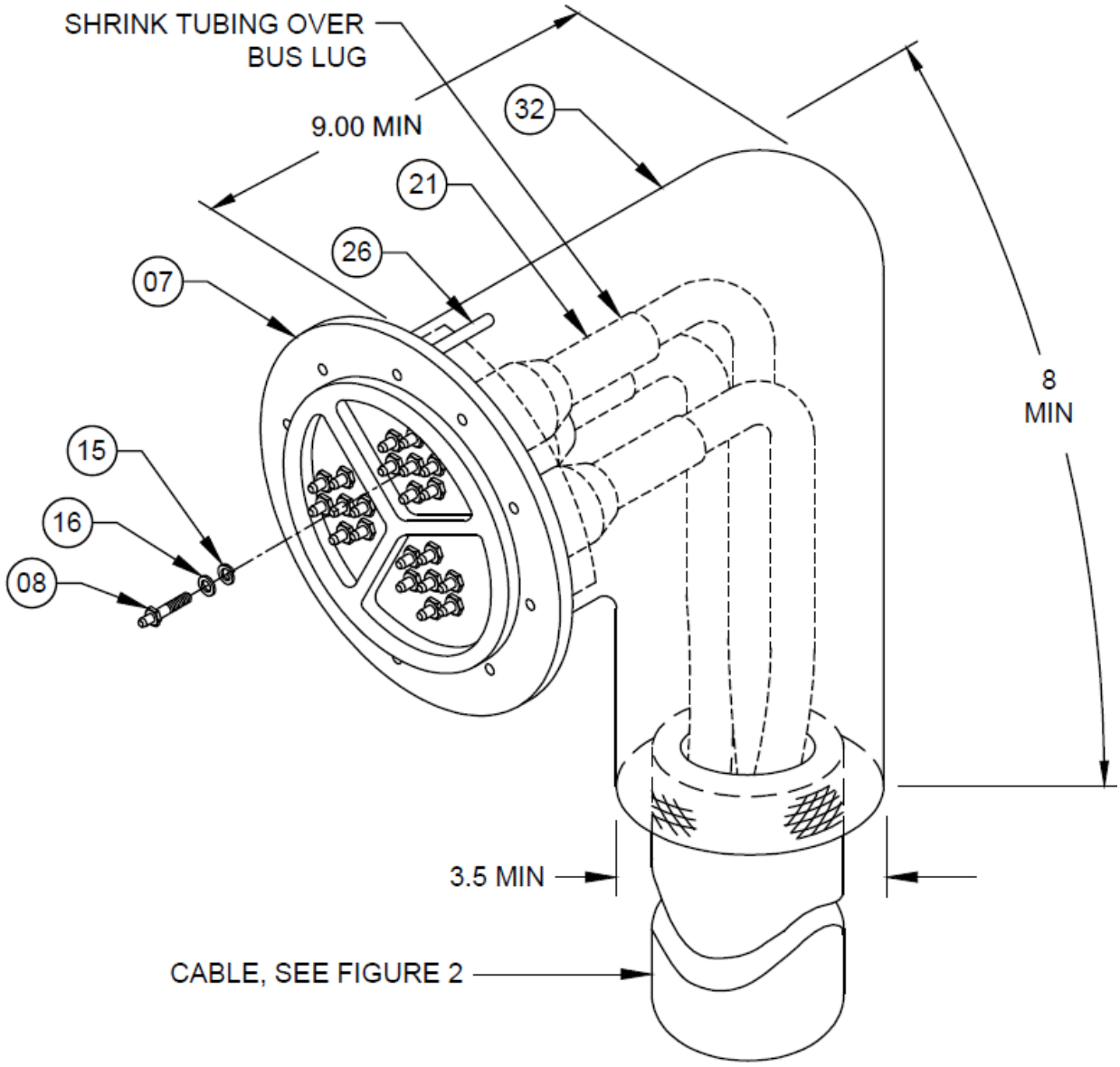
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NOTE: Cable termination shall be in accordance with MIL-STD-2003-1. The minimum cable bend radius, as specified by MIL-HDBK-299, shall not be exceeded. Potting shall be in accordance with potting requirements herein (see requirement 10).

FIGURE 8. Plug insert and cable assembly for straight receptacle, typical.

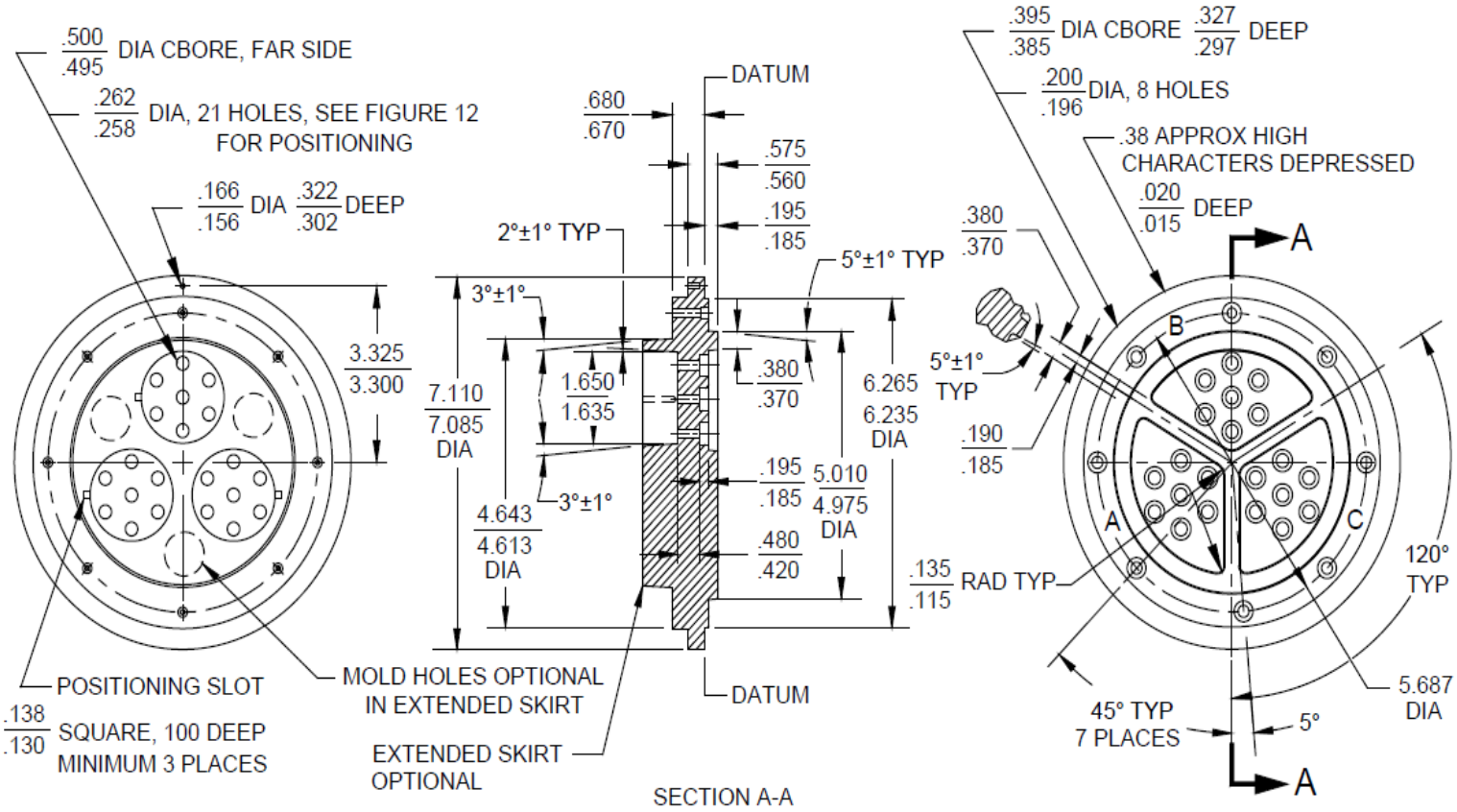
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NOTE: Cable termination shall be in accordance with MIL-STD-2003-1. The minimum cable bend radius, as specified by MIL-HDBK-299, shall not be exceeded. Potting shall be in accordance with potting requirements herein (see requirement 10).

FIGURE 9. Plug insert and cable assembly for 90-degree receptacle, typical.

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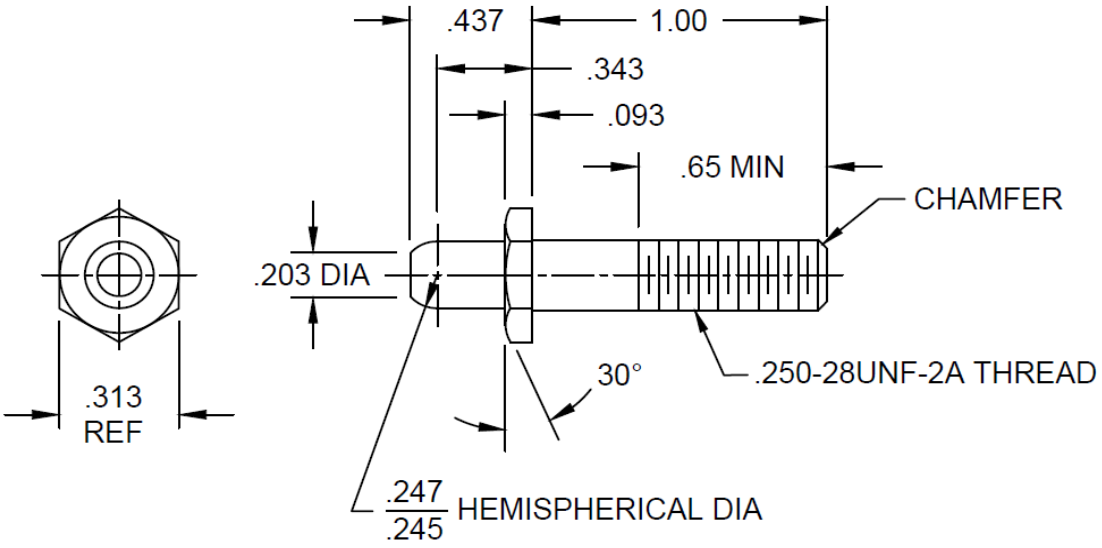


NOTES:

1. Areas in contact with the potting compound (see figures 8 and 9) and seal (figure 15) shall be sand blasted using 100 grit. Maximum removal of material shall be 0.001.
2. Apply enamel, commercial grade, to phase letters: A-Black (none), B-White, C-Red.

FIGURE 10. Insert, receptacle.

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NOTES:

1. All fillets corner radii shall be a maximum of 0.005.
2. Finish shall be $\sqrt{63}$ on all machined surfaces except as shown.
3. Pin shall be tellurium copper, UNS C14500 in accordance with ASTM B301/B301M. Silver plate shall be in accordance with ASTM B700, type 2, grade B, class S, 0.0002 to 0.0003 thick. A nickel or nickel alloy intermediate layer, at least 0.00004 thick, shall be applied before the silver electroplate.

FIGURE 11. Contact pin. (08)

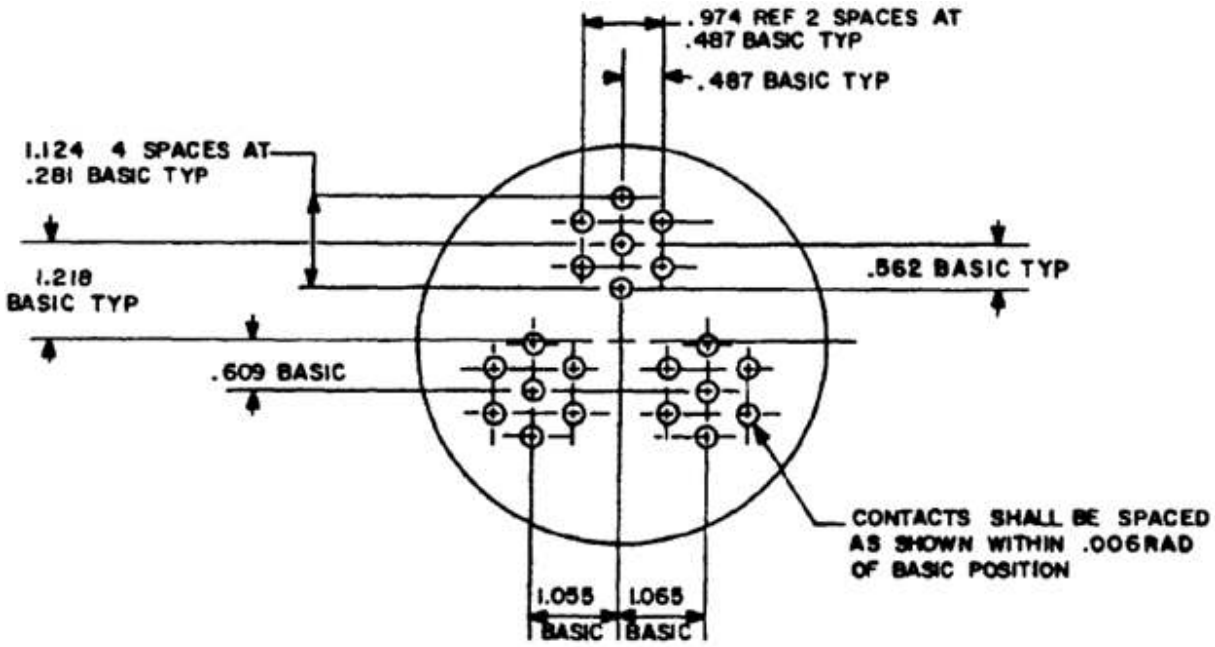


FIGURE 12. Positioning of contact centers.

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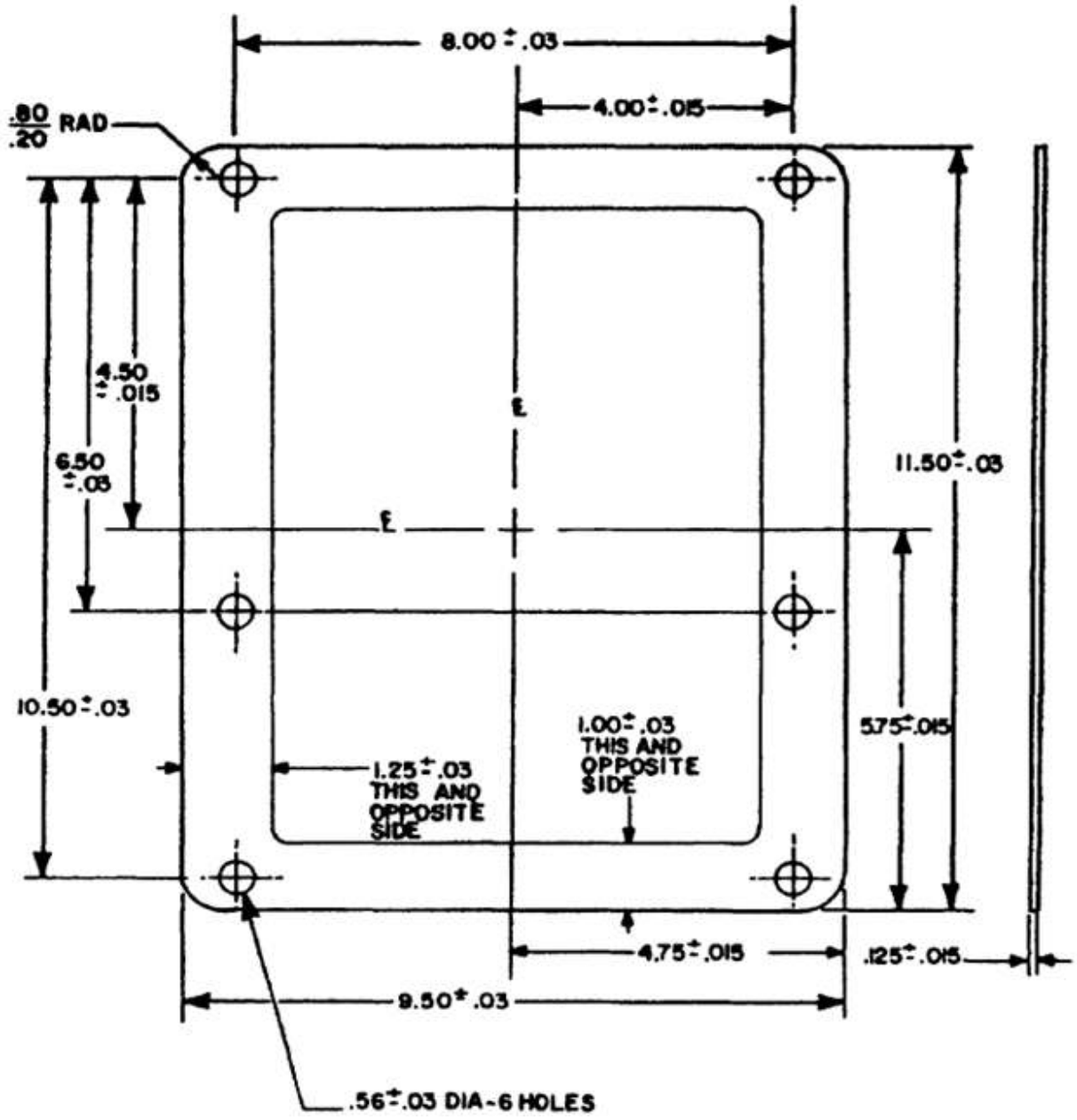


FIGURE 13. Gasket, receptacle. (09)

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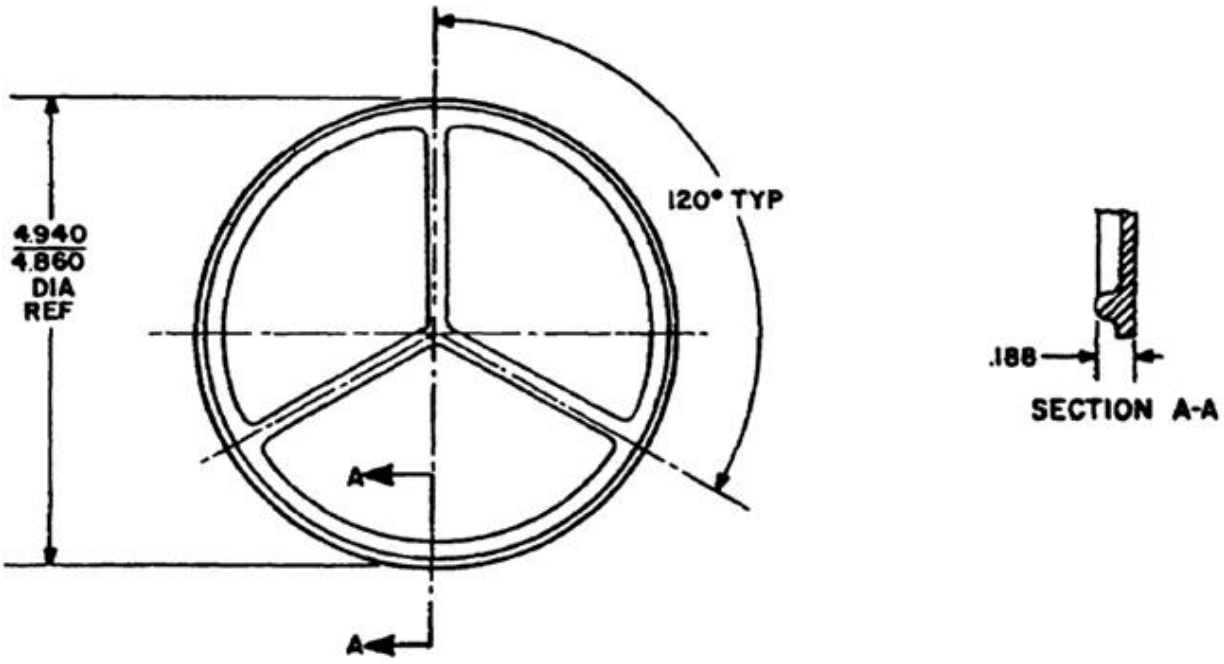


FIGURE 14 Seal, cover. (03)

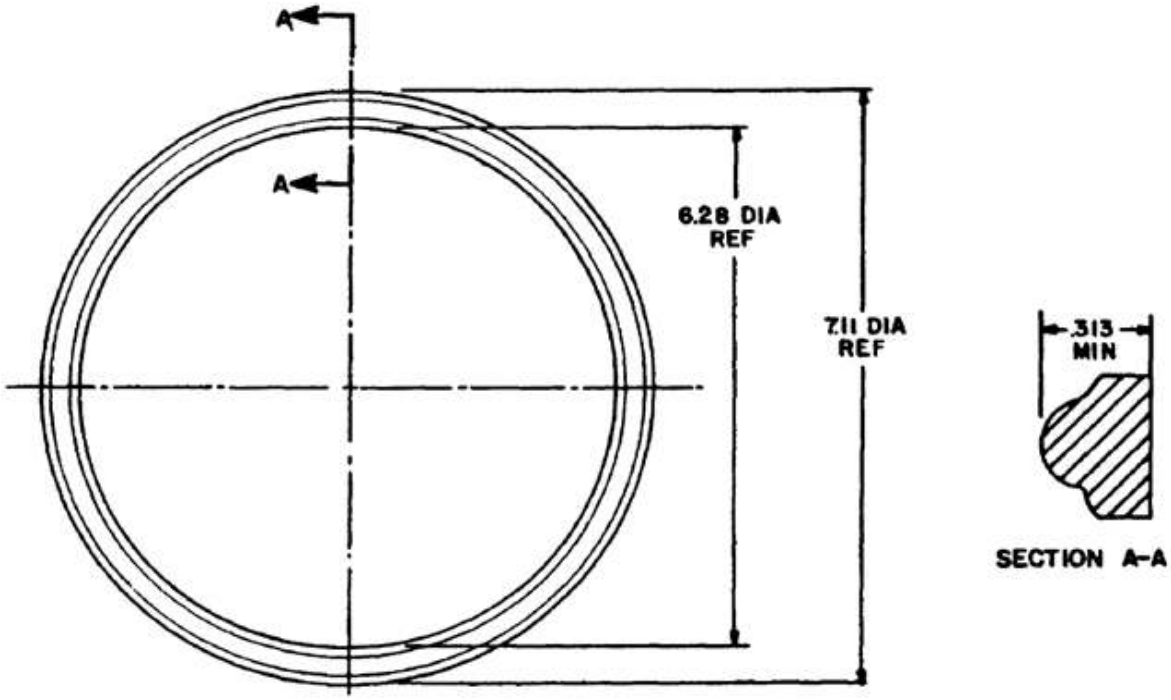
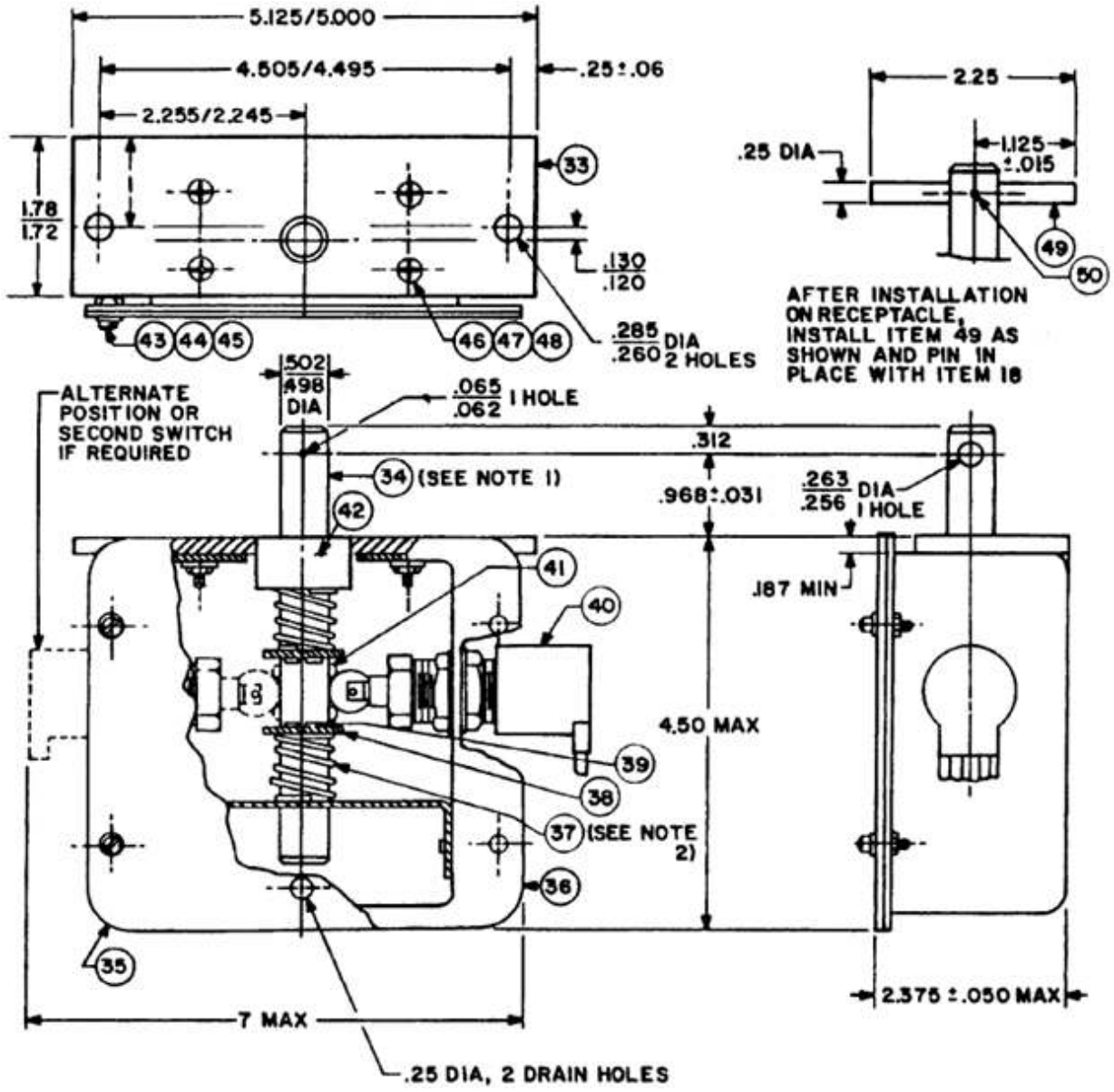


FIGURE 15. Seal, receptacle. (10)

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NOTES:

1. Finish of surface in contact with shaft seal shall be 32 microinches.
2. Spring shall have four full turns 0.50 inside diameter (id) minimum and 1-inch free length.

FIGURE 16. Switch housing assembly. (32)

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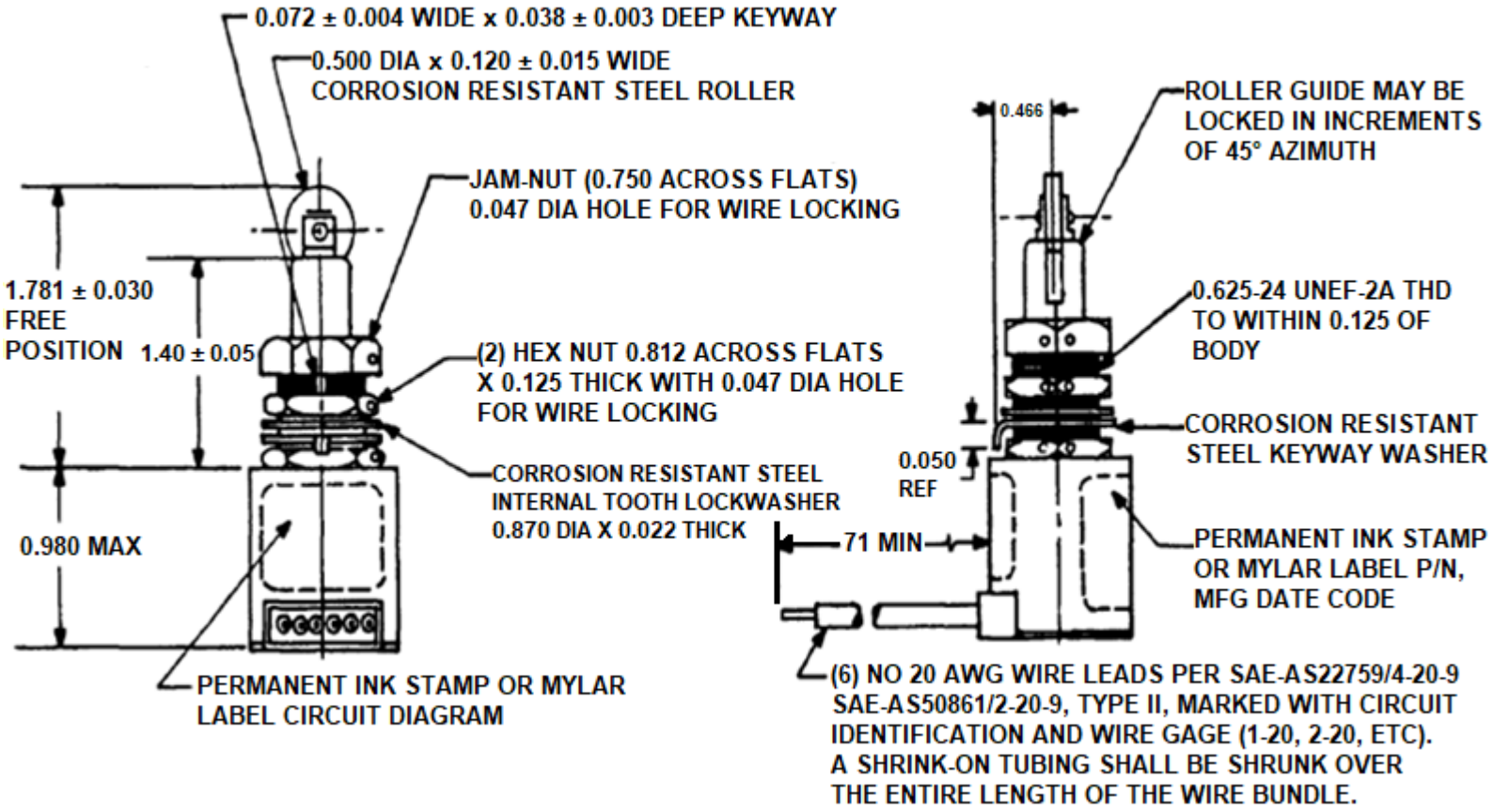


FIGURE 17. Switch. (40)

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CHARACTERISTICS:	
Contact arrangement	2 Single Pole Double Throw
Operating force°	9±3 pounds
Full overtravel force	30 pounds maximum
Release force	4 pounds minimum
Pretravel	.040 maximum
Movement differential	.020 maximum
Overtravel	.250 minimum
Break distance	.010 minimum
Mechanical and electrical endurance	See electrical rating
Weight with leads	.84 pounds maximum
Enclosure design	Watertight in accordance with enclosure design #3 of MIL-PRF-8805

ELECTRICAL RATING:
Switch shall be 450-volt, 2-ampere, 60-hertz, and shall pass an endurance test of 25,000 cycles at 450 volts, 2 amperes (inductive; 75-80 percent power factor), 60 hertz, 10 cycles per minute, on-off duration 10 percent on.

SOURCE OF SUPPLY:
Micro switch part no. 21EN-6, or equal.

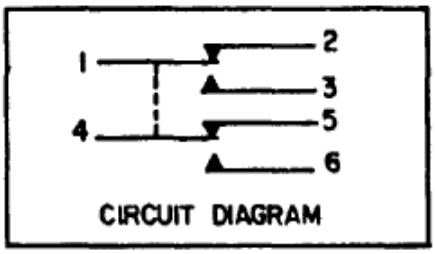


FIGURE 17. Switch. – Continued. (40)

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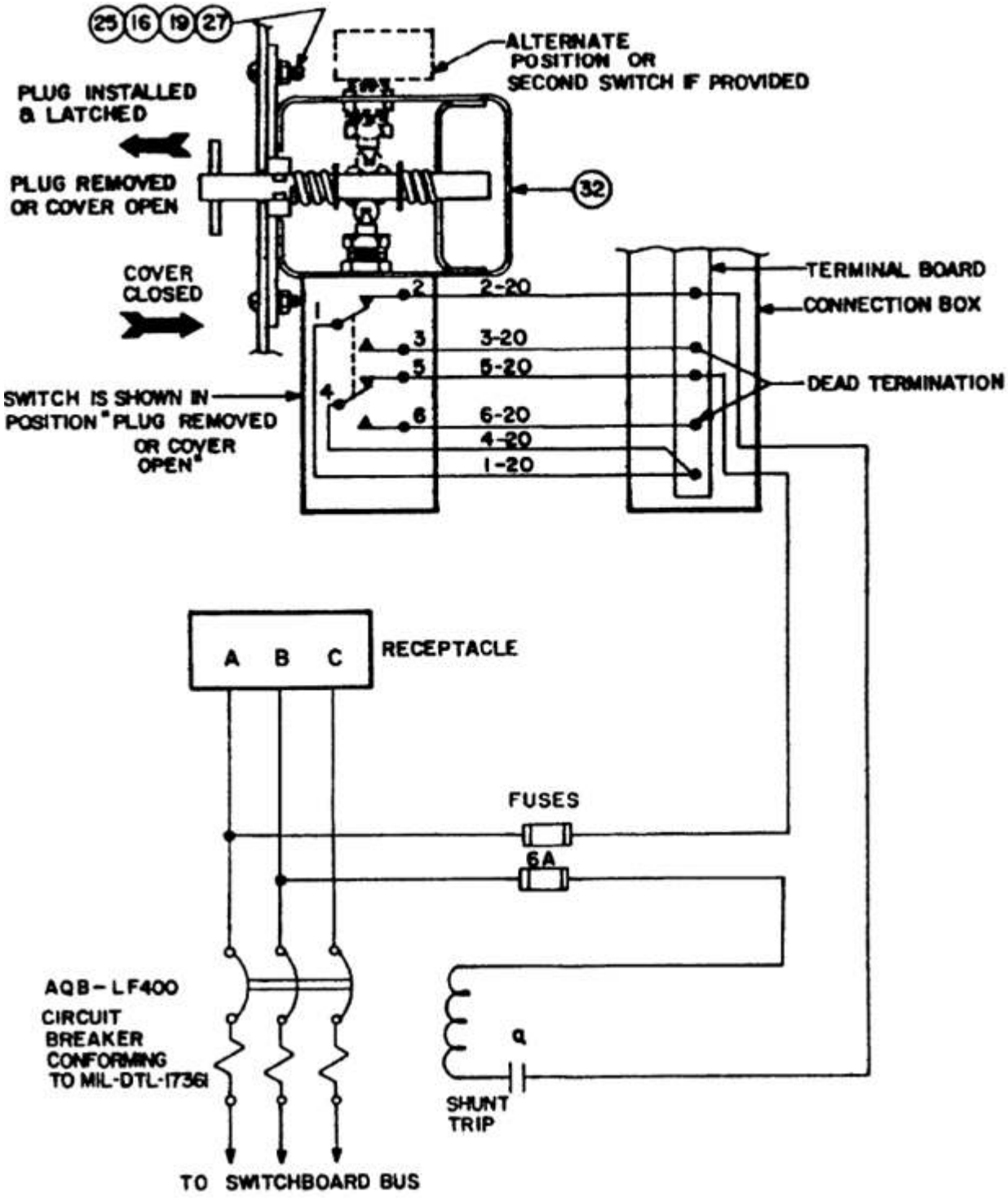


FIGURE 18. Typical wiring of one shipboard manually operated shore power circuit breaker.

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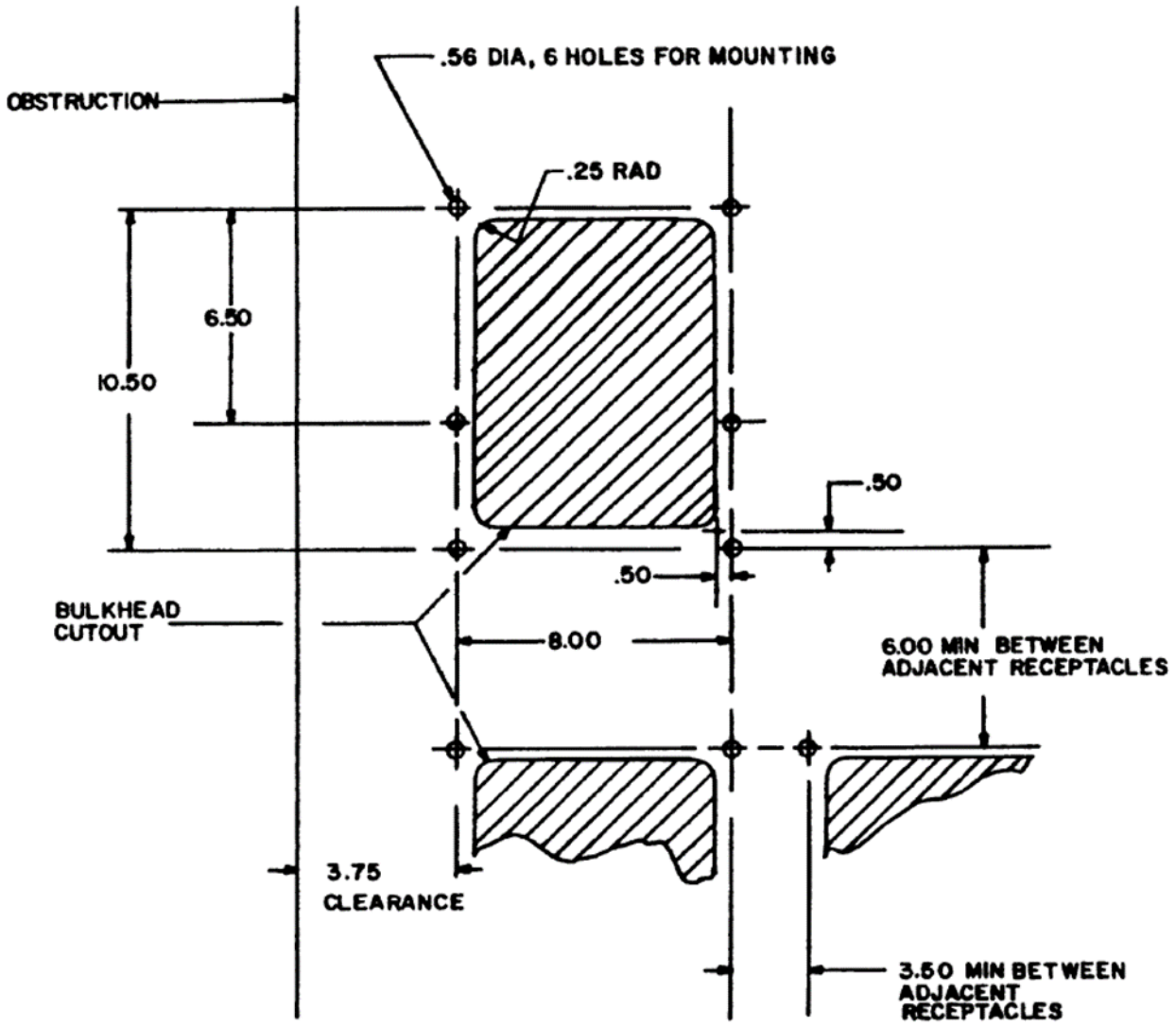


FIGURE 19. Mounting dimensions, bulkhead cutout and clearance.

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REQUIREMENTS:

1. Electrical rating: 500 volts, 500 amperes, 60 hertz. When 400 MCM cable is used, the receptacle rating is reduced to 400 amperes.

2. Dimensions and configuration: See figures 1 through 17.

3. Typical wiring: See figure 18.

4. Mounting dimensions: See figure 19.

5. Receptacle housing assembly: The receptacle shall be a unit having male contacts that is fixed on either the ship or shore installation. The receptacle housing shall provide a conical section for plug guiding, which shall protrude to protect the contact pins. A hinge point for the plug shall be provided. Provisions shall be made to replace the contact pins from the front of the receptacle, without removing the insert from the housing, for repairability. The cover shall be hinged to the receptacle housing and shall have provisions for mechanically locking the cover closed. When the plug is engaged and latched, or the cover is closed and latched, the switch shall be actuated.

6. Receptacle insert: The insert material shall be as shown in table I. Inserts shall have provisions for sealing between phases and external environments when the receptacle and plug are mated and latched or when the cover is closed and latched.

7. Contacts: The pin contact shall be mechanically and rigidly held in the insert. The contacts shall be in multiples per phase and terminated into a common bus bar per phase (see figures 10 and 12).

8. Cables: Cables shall be types 400 MCM or 500 MCM as shown in table II. Unless otherwise specified in the acquisition documents, cable length shall be 10 feet ±6 inches measured from the rear of the receptacle housing to the end of the cable. When receptacles are furnished without cable, the 500 MCM bus lugs shall have installed reducer inserts found in M24368/4. Receptacles with 400 MCM cables shall have reducer inserts installed in the 500 MCM bus lugs prior to crimping. Where single conductor cables are required, they shall be 400 MCM having the same characteristics as LSSHOF-500 cable, except for the rating. On the 90-degree receptacle, the cable outer jacket may be removed from the bend area but shall terminate inside the portion by not less than 1 inch.

9. Switch housing: Box sides and top shall have no other openings except for the mounting plate and switches. Switch openings shall polarize the switches from turning. A bracket shall be provided to hold and polarize the plunger to prevent it from turning more than 3 degrees. The switch shall operate (make or break) when the plunger moves a maximum of 0.25 inch in either direction from the rest position shown on figure 16. Parts shall work in unison without binding. The plunger shall positively return to the rest position after release of pressure or pull on the plunger. Normally switch housing assemblies with one switch are used. Switch housing assemblies with two switches are used for special wiring purposes and they shall be furnished only when specified in the acquisition documents. When only one switch is provided, a dust plug or cap shall be installed in the empty switch hole if required.

10. Potting: Potting procedures shall be in accordance with MIL-STD-2003-3. When a receptacle is to be provided with the cable (see table II), the potting shall be Loctite Stycast 2741BK with Stycast 2741 W1 – Catalyst 15, or equal.

11. Painting: Metal parts normally exposed to weather shall be painted. Painting shall be in accordance with MIL-DTL-917.

12. Standard parts: The standard parts shown in table I shall be used in the construction of the receptacle shown herein.

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TABLE I. Standard parts list.

Item no. 1/	Description	Quantity	Material/Specification	Remarks
01	Receptacle, housing	1	SS, 300 Series	figure 3
02	Cover, receptacle	1	SS, 300 Series	figure 4
03	Seal, cover	1	---	figure 14
04	Screw, set	1	NAS1091C4A12L	---
05	Latch	1	ASTM A666	figure 6
06	Bracket, latch locking	1	ASTM A666	figure 7
07	Insert, receptacle	1	2/	figure 10
08	Contact pin	21	---	figure 11
09	Gasket, receptacle	1	---	figure 13
10	Seal, receptacle	1	---	figure 15
11	Potting (straight 3 single condition)	½ gallon	See potting material requirements	---
12	Screw	8	NASM16996-15	---
13	Bolt	1	NAS1004-32A	---
14	Nut	1	NASM21043-4	---
15	Washer, flat	21	NAS620C416L	---
16	Washer, lock	23	NASM35338-139	---
17	Pin, cotter	2	NASM24665-302	---
18	Washer, flat	2	NAS1149C816L	---
19	Washer, flat	2	NAS1149C416L	---
20	Washer, flat	2	NAS1149C416	---
21	Bus lug, 500 MCM	3	M24368/1 Item 06	---
22	Washer, flat	8	NAS620C10L	---
23	Washer, lock	8	NASM35338-138	---
24	Nut, hex	8	NASM35650-304	---
25	Philips pan head screw (1/4-20 x 1.25 inches)	2	CRES, MS51957D	---
26	Guide pin	1	NASM16556-627	---
27	Nut	2	MS51972-1	---
28	Identification plate	1	3/	figure 5
29	Warning plate	1	3/	figure 2
30	Potting (straight, one 3 Conductor cable)	¾ gallon	See potting material requirements	figure 8
31	Potting (angled)	1¾ gallon	See potting material requirements	figure 9
32	Switch housing assembly	1	---	figure 16

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TABLE I. Standard parts list – Continued.

Item no. 1/	Description	Quantity	Material/Specification	Remarks
33	Mounting plate	1	CRES, ASTM A240/A240M or ASTM A666	---
34	Plunger	1	SS 304	---
35	Cover, box	1	CRES, ASTM A240/A240M or ASTM A666	---
36	Box, 0.062 thick	1	CRES, ASTM A240/A240M or ASTM A666	---
37	Spring, 0.062	1	ASTM A313/A313M	---
38	Washer, .875 od, .50 id, .093 thick	As Required	CRES, ASTM A240/A240M or ASTM A666	---
39	Pin, cotter	2	SS 304	---
40	Switch	1	---	figure 17
41	Rivet, 0.218 button or round head	1	SS 304	ASME B18.1.1
42	Shaft seal	1	---	4/
43	Screw	4	SS 304	---
44	Washer, spring lock	4	CRES, ASTM A240/A240M or ASTM A666	---
45	Nut, hex	4	SS 304	---
46	Screw	4	SS 304	---
47	Washer, spring lock	4	CRES, ASTM A240/A240M or ASTM A666	---
48	Nut	4	SS 304	---
49	Rod	1	SS 304	---
50	Pin, spring, 0.062 dia, 0.50 liter	1	CRES, commercial	---
51	Insert, reducer	3	M24368/4 Item 33	See cable requirement
52	Cable, LSSHOF-500	As Required	MIL-DTL-24643/3	---
53	Cable, LSTHOF-400/500	As Required	MIL-DTL-24643/3	---
54	Cable, LSTSGA-400	As Required	MIL-DTL-24643/16	---
55	Cable, LSTSGU-400	As Required	MIL-DTL-24643/16	---

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TABLE I. Standard parts list – Continued.

NOTES:
1/ Item numbers are the same as numbers shown in circles on figures.
2/ Molding material shall be glass roving reinforced phenolic, color black, Sumitomo GF 8202 L24, or equal.
3/ Attachment may be by adhesive backing or blind rivets.
4/ Shaft seal shall be lubricated with a light film of silicone compound conforming to SAE AMS8660 at the time of assembly.

13. High-impact shock test: N/A

GENERAL INFORMATION:

1. Part of Identifying Numbers (PINs) are created as shown in table II.

TABLE II. Assignment of PINs and general information.

PIN	Cable type (10 ft)	Former PIN
Straight receptacle – see figure 8		
M24368/2–001 thru 016	Discontinued	---
M24368/2–101	None	M24368/2–009 or 011
M24368/2–102	400 MCM 1/	M24368/2–001
M24368/2–103	LSSHOF-500 (3)	M24368/2–003
M24368/2–104	LSTHOF-400	M24368/2–001
M24368/2–105	LSTHOF-500	M24368/2–003
M24368/2–106	LSTSGU-400	M24368/2–001
M24368/2–107	LSTSGA-400	M24368/2–003
90-degree receptacle – see figure 9		
M24368/2–201	400 MCM 1/	NONE
M24368/2–202	LSSHOF-500	NONE
M24368/2–203	LSTHOF-400	NONE
M24368/2–204	LSTHOF-500	NONE
M24368/2–205	LSTSGA-400	NONE
M24368/2–206	LSTSGU-400	NONE

NOTE:
1/ Three single conductor cables.

2. Typical installation of shore power receptacle is shown in MIL-STD-2003-2 and MIL-STD-2003-3.

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3. Contact information for the part numbers specified herein is as follows:

Honeywell Corp. (CAGE 91929) https://honeywell.com (800) 537-6945 1985 Douglas Drive North Golden Valley, MN 55422	Henkel Corp. (CAGE 80UG0) https://www.henkel.com (908) 685-7000 10 Finderne Avenue Bridgewater, NJ 08807	Sumitomo Bakelite North America, Inc. (CAGE 1JX75) https://www.sbhpp.com (248) 313-7000 46820 Magellan Drive, Suite C Novi, MI 48377

ACQUISITION REQUIREMENTS:

1. Length shall be specified when more than 10 feet ±6 inches of cable is required.
2. Two switches shall be specified when required.

CHANGES FROM PREVIOUS ISSUE Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extent of the changes.

CONCLUDING MATERIAL

Custodian:
Navy – SH

Review activity:
Navy – EC

Preparing activity:
Navy − SH
(Project 6150-2020-002)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil.

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